

ELLISTON, Australia

WMO: 946560

Lat: 33.650S

Lon: 134.888E

Elev: 9

StdP: 101.22

Time Zone: 9.50 (AUA)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	5.9	0.3	3.9	16.1	1.9	4.3	13.5	13.0	13.0	11.3	13.6	0.6	90	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	34.0	20.2	31.3	19.4	29.0	18.4	22.2	29.6	20.9	27.0	19.9	25.7	4.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	14.8	24.2	18.8	13.6	23.0	18.0	13.0	22.2	65.1	29.9	60.5	27.2	57.2	25.7	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.4	8.5	1.9	39.0	1.6	2.0	0.7	40.4	-0.2	41.6	-1.2	42.7	-2.4	44.1		
(5)			1.5	24.8	1.6	3.1	0.4	27.0	-0.5	28.8	-1.4	30.5	-2.5	32.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	20.9	20.5	19.0	17.1	14.8	13.0	12.3	13.0	14.6	16.1	18.0	19.4		
	DBStd	4.13	3.70	3.34	3.23	2.90	2.27	2.04	1.85	2.04	2.68	3.26	3.56	3.52		
	HDD10.0	6	0	0	0	0	0	2	3	1	0	0	0	0		
	HDD18.3	999	6	6	26	58	113	161	187	166	119	89	46	24		
	CDD10.0	2389	337	295	278	212	149	92	74	94	137	189	242	290		
	CDD18.3	339	84	67	45	20	4	0	0	1	6	19	37	56		
	CDH23.3	2664	698	462	328	111	19	1	0	3	57	187	330	467		
	CDH26.7	1061	328	193	116	29	4	0	0	0	11	54	129	197		
(14) Wind	WSAvg	4.0	4.9	4.7	3.9	3.1	2.9	3.2	3.4	3.7	4.0	4.4	4.7	4.8		
(15) Precipitation	PrecAvg	429	11	14	16	27	54	65	70	62	44	30	18	17		
	PrecMax	646	46	100	61	132	146	150	134	116	126	97	60	66		
	PrecMin	229	0	0	0	0	13	27	21	13	6	3	0	0		
	PrecStd	93	10	18	18	26	30	27	30	28	28	22	13	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	35.7	34.2	29.8	25.5	21.1	20.4	23.1	28.2	31.5	34.6	35.5		
		MCWB	21.5	22.0	19.7	17.2	15.1	14.0	13.1	14.2	16.5	16.9	19.1	20.4		
	2%	DB	33.8	32.2	29.8	26.3	22.1	18.5	18.1	20.3	24.7	27.7	30.4	31.5		
		MCWB	20.2	21.0	18.7	17.0	14.7	13.7	12.8	13.4	15.8	15.6	18.3	19.1		
	5%	DB	30.0	29.3	27.4	23.8	20.0	17.2	16.6	18.1	22.1	24.9	27.2	28.8		
		MCWB	19.6	19.8	17.9	16.5	15.5	13.4	12.7	12.8	14.9	15.2	17.2	18.1		
	10%	DB	26.7	26.1	24.6	21.6	18.6	16.4	15.7	16.6	19.4	21.9	24.1	25.6		
		MCWB	19.1	19.0	17.7	16.1	14.8	13.1	12.3	12.5	14.0	14.5	16.8	17.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.9	24.7	22.1	20.5	18.2	16.1	14.9	15.8	19.1	18.6	21.1	22.2		
		MCDB	31.0	31.1	28.8	24.9	20.5	19.0	17.2	20.3	24.9	26.6	28.9	29.7		
	2%	WB	22.0	22.1	19.9	18.6	17.1	15.0	14.1	14.6	17.0	17.2	19.5	20.6		
		MCDB	29.9	29.7	26.3	23.0	19.4	17.0	16.3	17.7	21.8	23.6	26.8	26.8		
	5%	WB	20.6	20.8	19.1	17.7	16.1	14.3	13.5	13.8	15.8	16.3	18.5	19.6		
		MCDB	26.7	26.5	24.0	21.3	18.8	16.3	15.7	16.8	19.9	21.6	24.6	25.7		
	10%	WB	19.6	19.7	18.3	17.0	15.3	13.7	12.9	13.2	14.9	15.5	17.5	18.6		
		MCDB	25.0	24.4	23.1	20.4	18.0	15.8	15.2	16.0	18.5	20.1	22.9	24.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	7.3	8.0	8.6	7.7	7.1	7.2	7.9	8.5	9.3	8.6	8.6		
		MCDBR	13.4	12.3	12.1	11.2	9.7	8.1	8.8	11.3	13.8	14.6	13.5	13.9		
		MCWBR	5.5	5.0	5.3	6.0	6.0	5.2	5.7	6.8	7.4	6.9	5.8	5.8		
	5% WB	MCDBR	10.3	9.2	9.6	9.5	8.1	6.8	7.2	9.1	11.2	11.8	10.7	10.2		
		MCWBR	5.6	4.6	5.0	6.3	6.1	5.0	5.5	6.4	7.2	6.7	5.9	5.6		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.347	0.329	0.321	0.310	0.303	0.301	0.308	0.335	0.343	0.345	0.338		
	taud		2.556	2.565	2.601	2.614	2.624	2.622	2.627	2.592	2.505	2.484	2.510	2.558		
	Ebn at Noon		992	970	953	907	865	844	867	907	930	962	986	1003		
	Edh at Noon		108	103	93	82	72	67	70	82	100	111	112	108		
(44) All-Sky Solar Radiation	RadAvg		7.84	6.97	5.67	4.17	2.97	2.50	2.71	3.65	4.95	6.25	7.20	7.75		
	RadStd		0.35	0.36	0.28	0.26	0.13	0.12	0.13	0.22	0.33	0.40	0.39	0.39		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page